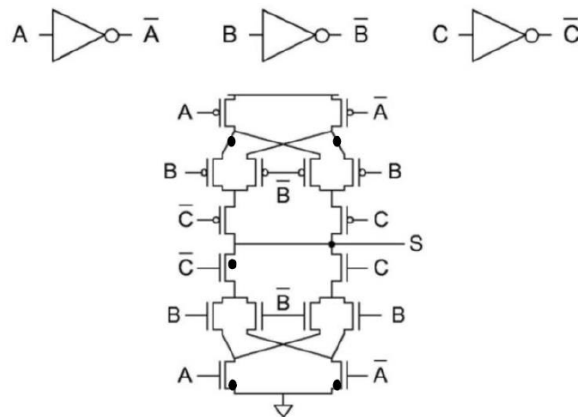


1. The circuit diagram of your design and explaining your design (You can use screenshots to explain)



完全照著上圖做的

```
.subckt Inv in1 inv_out VDD GND
mp1 inv_out in1 VDD VDD P_18 w=1.2u l=0.18u
mn1 inv_out in1 GND GND N_18 w=0.6u l=0.18u
.ends
```

Inverter

```
.subckt Xor A B C S VDD GND
** inv
Xinv1 A A_inv VDD GND Inv
Xinv2 B B_inv VDD GND Inv
Xinv3 C C_inv VDD GND Inv
** pmos
mp1 P_A_inv_out A_inv VDD VDD P_18 w=1.2u l=0.18u
mp2 P_A_out A VDD VDD P_18 w=1.2u l=0.18u
mp3 P_B_out_L B P_A_out VDD P_18 w=1.2u l=0.18u
mp4 P_B_out_R B P_A_inv_out VDD P_18 w=1.2u l=0.18u
mp5 P_B_out_L B_inv P_A_inv_out VDD P_18 w=1.2u l=0.18u
mp6 P_B_out_R B_inv P_A_out VDD P_18 w=1.2u l=0.18u
mp7 S C_inv P_B_out_L VDD P_18 w=1.2u l=0.18u
mp8 S C P_B_out_R VDD P_18 w=1.2u l=0.18u
** nmos
mn1 N_A_out A GND GND N_18 w=0.6u l=0.18u
mn2 N_A_inv_out A_inv GND GND N_18 w=0.6u l=0.18u
mn3 N_B_out_L B N_A_out GND N_18 w=0.6u l=0.18u
mn4 N_B_out_R B N_A_inv_out GND N_18 w=0.6u l=0.18u
mn5 N_B_out_L B_inv N_A_inv_out GND N_18 w=0.6u l=0.18u
mn6 N_B_out_R B_inv N_A_out GND N_18 w=0.6u l=0.18u
mn7 S C_inv N_B_out_L GND N_18 w=0.6u l=0.18u
mn8 S C N_B_out_R GND N_18 w=0.6u l=0.18u
.ends
```

3-input XOR

這段 3-input XOR 的 code 由上往下可以分成三個部分:

1. 先把 ABC 都先做 Inv，之後會用到
2. Pmos 區，這邊把八個 Pmos 都做出來，也就是上面的電路圖上方區域的邏輯
3. Nmos 區，這邊把八個 Nmos 都做出來，也就是上面的電路圖下方區域的邏輯

基本上只是照著電路圖慢慢刻，沒有什麼特別的地方

結果:

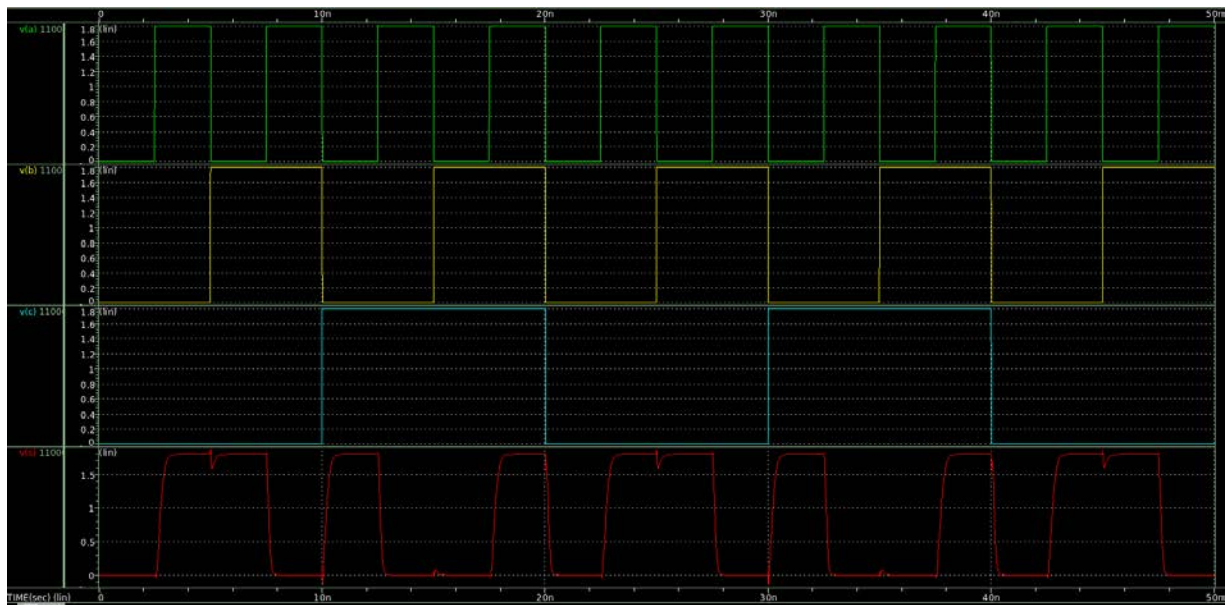
```
*****
** 110062222_hw2 **

***** transient analysis tnom= 25.000 temp= 25.000 *****
power= 22.3896u from= 0. to= 50.0000n
delay1_xor= 148.7402p targ= 20.1637n trig= 20.0150n
delay2_xor= 255.3041p targ= 2.7603n trig= 2.5050n

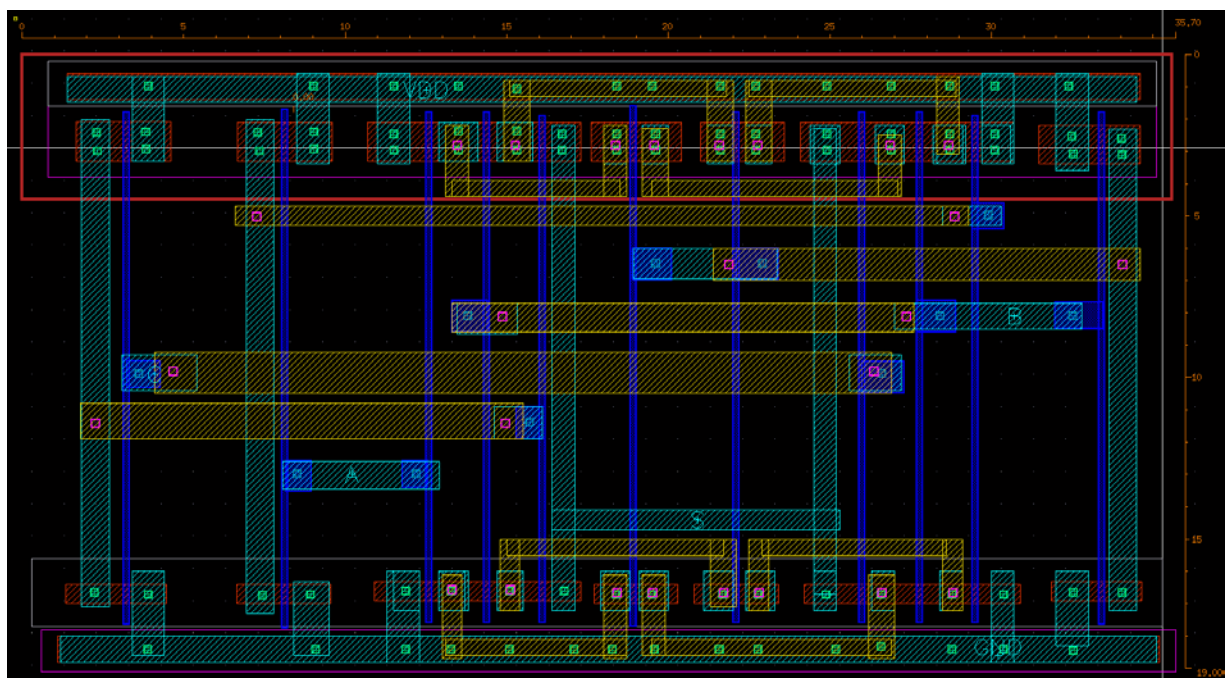
***** job concluded
*****
** 110062222_hw2 **

***** job statistics summary tnom= 25.000 temp= 25.000 *****
```

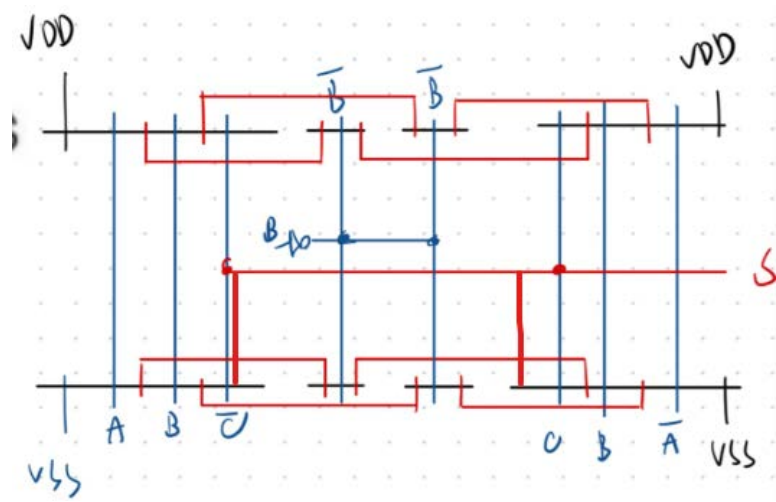
2. Pre-sim waveform



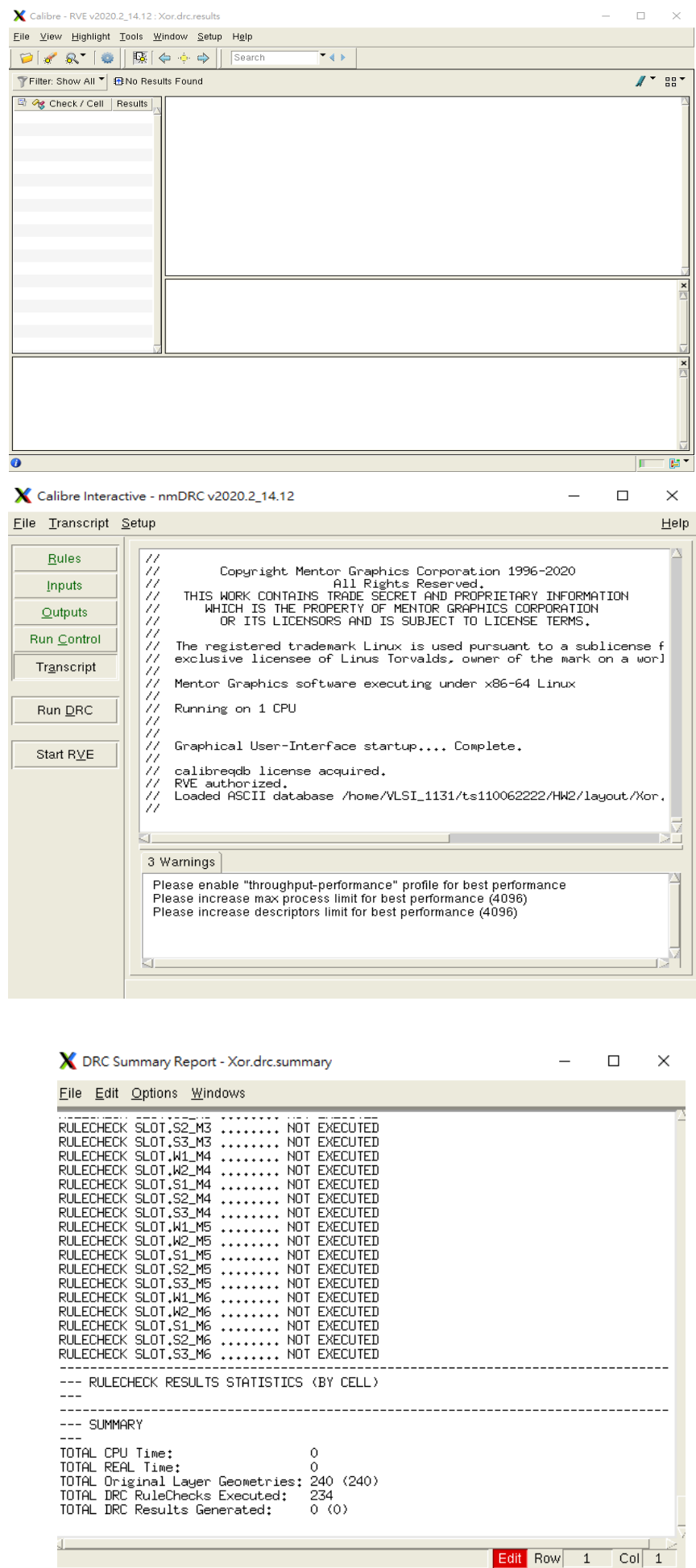
3. Screenshot of your layout (need to show the total area measurement with ruler in Virtuoso)



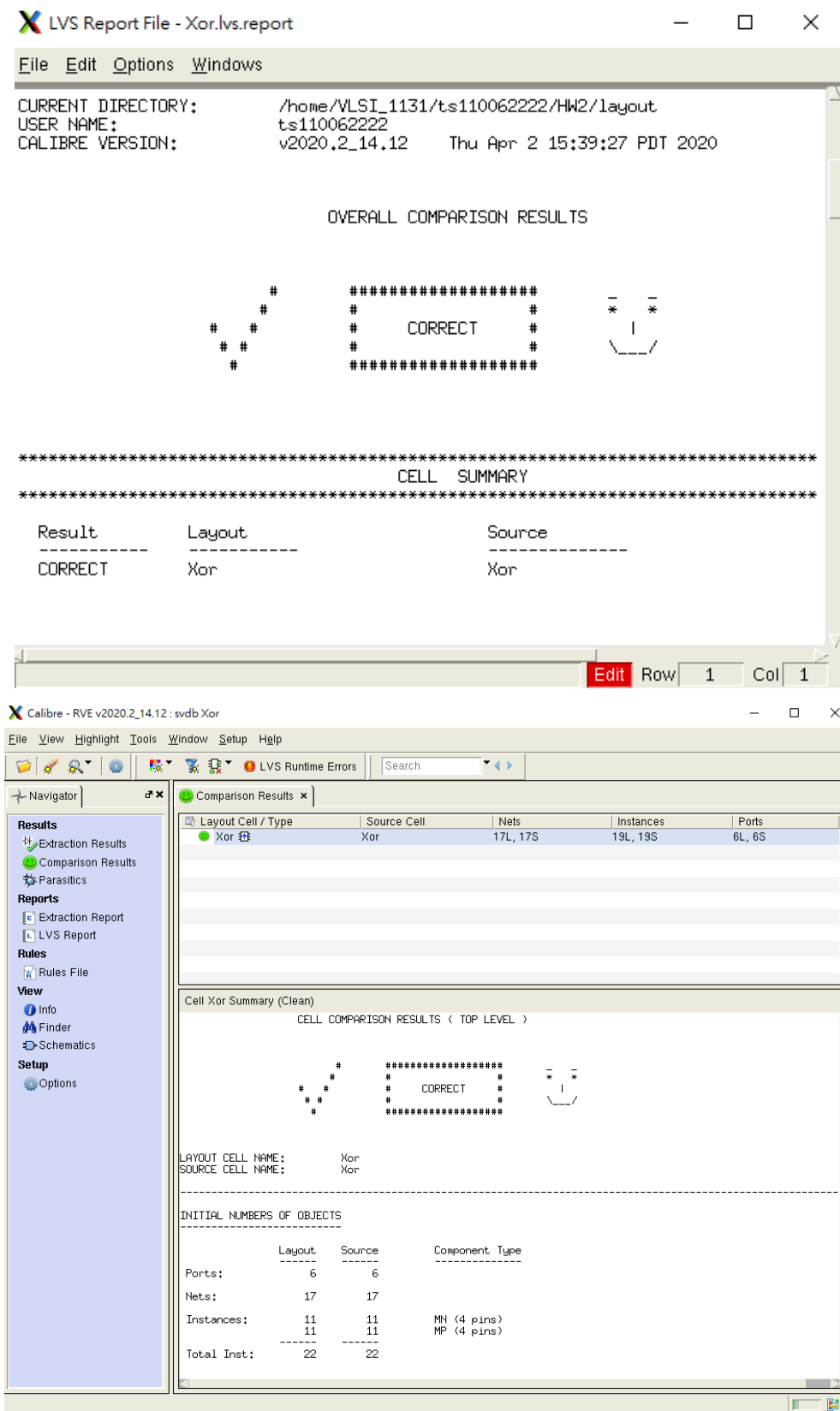
大概參考這個圖



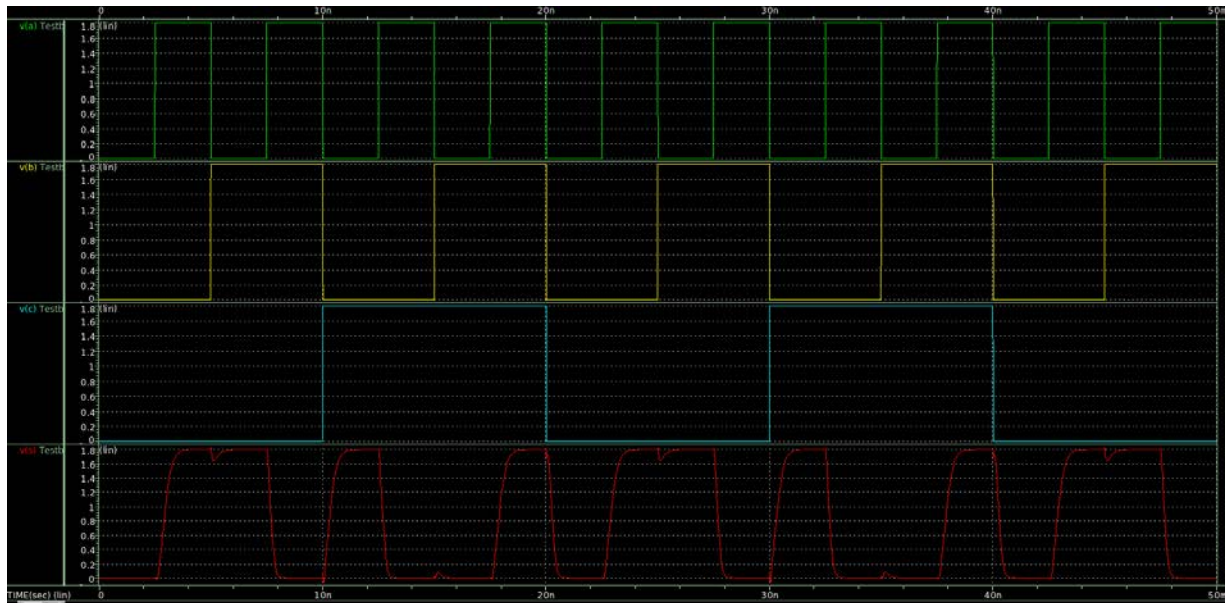
4. Screenshot of DRC summary report



5. Screenshot of LVS report includes the message of passing LVS



6. Post-sim waveform



7. Screenshot of the post-simulation result

```

*****
.protect

***** transient analysis tnom= 25.000 temp= 25.000 *****
power= 40.7200u from= 0. to= 50.0000n
delay1_xor= 265.4899p targ= 20.2805n trig= 20.0150n
delay2_xor= 437.3023p targ= 2.9423n trig= 2.5050n

***** job concluded

*****
.protect

***** job statistics summary tnom= 25.000 temp= 25.000 *****

```

8. Write down your delay and the difference in delay between rising and falling delay in the 3-input XOR gate.

Pre-sim

Delay1_{XOR}=__148.7402__ ps

Delay2_{XOR}=__255.3041__ ps

|Delay1_{XOR}–Delay2_{XOR}| =__106.5639__ ps

Post-sim

Delay1_{XOR}=__265.4899__ ps

Delay2_{XOR}=__437.3023__ ps

|Delay1_{XOR}–Delay2_{XOR}| =__171.8124__ ps

9. The hardness of this assignment and how you overcome it.

The hardness of this assignment.

最難的絕對是畫 layout，畫到眼睛快脫窗，可能我 stick diagram 畫的不好吧，第一次接觸這東西就要畫這麼複雜的東東，很怕畫半天結果走錯路

How I overcome it?

土法煉鋼，做的出來比較重要，盡量畫大一點可以忽略某些 design rule

10. Any suggestions about this programming assignment

感覺電路圖題目可以再簡單一點 qq，好多東西都摸好久才知道怎麼用，但討論區的助教甚麼都會，我遇到很多問題都在那邊解決的，感覺很棒